ASSESSMENT TOOL 2

Scenario-Based Research Assessment

CO1_AT2

MLA 0104 AI and Expert System

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DATE OF SUBMISSION

21-07-2026

ACKNOWLEDGEMENT

I express my sincere gratitude to our respected Principal, Head of the Department, and all the faculty members of the Department of Artificial Intelligence and Machine Learning for their continuous support and encouragement in completing this assignment.

I would like to express my heartfelt thanks to my faculty guide for providing valuable guidance, constructive suggestions, and constant motivation throughout the preparation of this report. Their support helped me understand the concepts of Artificial Intelligence, Autonomous Vehicles, Computer Vision, and Intelligent Transportation Systems.

I also thank **Waymo** and the developers of autonomous vehicle technologies for publishing technical information, research papers, and documentation that greatly helped in understanding the working mechanism of AI-powered self-driving vehicles.

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Finally, I express my sincere gratitude to my parents, friends, and classmates for their encouragement and support throughout the completion of this assignment.

ABSTRACT

Artificial Intelligence (AI) is transforming the transportation industry by enabling vehicles to operate with minimal or no human intervention. Autonomous vehicles are one of the most significant applications of AI, combining technologies such as Machine Learning, Deep Learning, Computer Vision, Sensor Fusion, and Intelligent Decision-Making to understand road conditions, detect obstacles, predict traffic behavior, and navigate safely. These intelligent systems aim to improve road safety, reduce accidents caused by human error, and enhance transportation efficiency.

This report focuses on the 6th-Generation Waymo Driver, an advanced autonomous driving system developed by Waymo**.** The system integrates high-resolution cameras, LiDAR, radar, GPS, and AI-based software to perceive the surrounding environment and make real-time driving decisions. It continuously processes sensor data, identifies nearby vehicles, pedestrians, cyclists, traffic signals, and road signs, predicts their movements, and plans the safest driving path. The combination of advanced sensing technologies and AI enables the vehicle to operate efficiently under various traffic conditions.

The report presents the problem addressed by autonomous vehicles, the need for Artificial Intelligence, AI technologies involved, working mechanism, data processing techniques, real-world implementation, benefits, limitations, and future developments. It also discusses the impact of AI on modern transportation systems and highlights the importance of intelligent autonomous driving in building safer and smarter cities.

Overall, the 6th-Generation Waymo Driver demonstrates how Artificial Intelligence can transform the transportation industry through intelligent perception, decision-making, and autonomous navigation. Continuous research and technological advancements are expected to further improve the safety, reliability, and large-scale adoption of autonomous vehicles in the future.

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TITLE OF AI APPLICATION

6th-Generation Waymo Driver: An AI-Powered Autonomous Vehicle System

AI Domain

Autonomous Vehicles

The selected AI application for this report is the **6th-Generation Waymo Driver**, an advanced autonomous driving system developed by **Waymo**, a leader in self-driving vehicle technology. The Waymo Driver combines Artificial Intelligence, advanced sensors, and intelligent software to enable vehicles to navigate roads safely without continuous human control. It is designed to understand the surrounding environment, detect objects, predict the behavior of other road users, and make safe driving decisions in real time.

The application belongs to the **Autonomous Vehicles** domain because it performs intelligent driving tasks such as environment perception, object detection, lane navigation, traffic sign recognition, obstacle avoidance, and route planning using AI algorithms. The Waymo Driver represents one of the most advanced examples of autonomous driving technology currently operating in real-world transportation.

DOMAIN SELECTION

Selected Domain: Autonomous Vehicles

Autonomous vehicles are intelligent transportation systems that use Artificial Intelligence to perform driving tasks traditionally carried out by human drivers. These vehicles are equipped with multiple sensors, including cameras, LiDAR, radar, GPS, and ultrasonic sensors, which continuously monitor the surrounding environment. AI algorithms analyze this information to recognize objects, understand traffic conditions, predict possible hazards, and control vehicle movement safely.

The **6th-Generation Waymo Driver** is an ideal example of the Autonomous Vehicles domain because it integrates several AI technologies to achieve safe and reliable autonomous driving. By reducing dependence on human drivers, autonomous vehicles have the potential to improve road safety, increase transportation efficiency, reduce traffic congestion, and support the development of intelligent transportation systems.

INTRODUCTION AND BACKGROUND

Transportation is one of the most important sectors supporting economic growth and daily life. Every day, millions of people and goods are transported across cities and countries using road networks. However, traditional transportation systems rely heavily on human drivers, making them vulnerable to challenges such as driver fatigue, distraction, human error, traffic congestion, and road accidents. According to road safety studies, human error is a major contributing factor in many traffic accidents, highlighting the need for safer and more intelligent transportation solutions.

Artificial Intelligence has introduced significant advancements in the field of autonomous vehicles. Modern self-driving vehicles use AI to observe their surroundings, understand complex traffic situations, predict the behavior of nearby vehicles and pedestrians, and make driving decisions without constant human intervention. These capabilities allow autonomous vehicles to improve driving safety, optimize traffic flow, and provide reliable transportation services.

The 6th-Generation Waymo Driver, developed by Waymo, is one of the latest and most advanced autonomous driving systems. It combines Computer Vision, Deep Learning, Machine Learning, Sensor Fusion, Localization, and Path Planning to navigate roads safely. The system collects information using cameras, LiDAR, radar, GPS, and other sensors. Artificial Intelligence algorithms process this information to detect lanes, traffic signs, vehicles, pedestrians, cyclists, and obstacles while continuously planning safe driving actions.

Unlike conventional vehicles, the Waymo Driver constantly updates its understanding of the surrounding environment. It predicts possible movements of nearby road users, selects the safest route, controls steering, braking, and acceleration, and continuously monitors traffic conditions. These intelligent capabilities enable the vehicle to operate in complex urban environments while maintaining a high level of safety and reliability.

The development of autonomous driving technology represents a major milestone in the evolution of intelligent transportation systems. By reducing human intervention and improving decision-making through AI, autonomous vehicles have the potential to transform the future of mobility. They can contribute to safer roads, improved logistics, efficient public transportation, and smarter cities. As AI technology continues to evolve, autonomous driving systems such as the 6th-Generation Waymo Driver are expected to play an increasingly important role in the future of transportation.

PROBLEM IDENTIFICATION

The transportation industry faces several challenges that affect road safety, traffic efficiency, and logistics operations. Traditional vehicles depend entirely on human drivers, making transportation vulnerable to human errors such as fatigue, distraction, speeding, poor decision-making, and violation of traffic rules. These factors contribute to a significant number of road accidents and traffic congestion worldwide.

Another major challenge is the increasing demand for safe, efficient, and reliable transportation services. As the number of vehicles on roads continues to grow, managing traffic and ensuring smooth transportation becomes more difficult. Long-distance driving also places physical and mental stress on drivers, increasing the risk of accidents during extended journeys.

Conventional transportation systems have limited capability to continuously monitor the surrounding environment and react instantly to unexpected situations. Human drivers may fail to detect obstacles, pedestrians, cyclists, or sudden changes in traffic conditions. These limitations highlight the need for intelligent systems that can perceive the environment accurately and make quick decisions.

The **6th-Generation Waymo Driver** addresses these challenges by using Artificial Intelligence to continuously observe road conditions, recognize surrounding objects, predict the behavior of other road users, and make safe driving decisions in real time. The system aims to improve road safety, reduce human errors, enhance transportation efficiency, and support the development of intelligent autonomous vehicles.

Figure 1: Problem Identification

Figure 1: Severe Traffic Congestion and Road Accident (The Problem).

NEED FOR AI SOLUTION

Artificial Intelligence plays a crucial role in autonomous vehicles because traditional driving methods cannot efficiently process the large amount of information generated during real-world driving. A human driver has limitations in continuously monitoring every surrounding object and reacting instantly to changing traffic conditions. AI overcomes these limitations by analyzing data from multiple sensors simultaneously and making accurate driving decisions within milliseconds.

The **6th-Generation Waymo Driver** uses AI to identify vehicles, pedestrians, traffic signs, road markings, and obstacles while continuously predicting their future movements. Machine Learning and Deep Learning algorithms enable the system to learn from large datasets and improve its driving performance over time. Computer Vision helps the vehicle understand visual information captured by cameras, while Sensor Fusion combines data from cameras, LiDAR, radar, and GPS to create a complete understanding of the surrounding environment.

By using Artificial Intelligence, autonomous vehicles can respond more consistently to complex road situations, reduce the possibility of human error, improve traffic flow, and provide safer transportation. AI also enables continuous monitoring, faster decision-making, and efficient route planning, making autonomous driving more reliable than traditional approaches in many driving scenarios

Figure 2: Need for AI Solution

Figure 2: Waymo Autonomous Vehicle Navigating Efficiently (The Need for AI Solution).

AI TECHNOLOGIES AND PRODUCTS USED

6.1 Waymo Driver (Main Autonomous Driving System)

The **Waymo Driver** is the core autonomous driving system developed by Waymo. It combines Artificial Intelligence, advanced sensors, and onboard computing to drive vehicles without continuous human intervention. The system can detect objects, understand road conditions, predict the behavior of nearby vehicles and pedestrians, and make safe driving decisions in real time. It is designed to improve road safety, reduce human error, and provide efficient transportation.

6.2 LiDAR Sensor

LiDAR (Light Detection and Ranging) is one of the most important sensors used in the Waymo Driver. It emits laser beams to create a detailed three-dimensional map of the surrounding environment. LiDAR accurately measures the distance, shape, and position of nearby objects, allowing the vehicle to identify roads, vehicles, pedestrians, cyclists, and obstacles even in low-light conditions. This information is essential for safe navigation and obstacle avoidance.

6.3 Camera System

The Waymo Driver uses multiple high-resolution cameras to capture visual information from all directions. The cameras detect lane markings, traffic lights, road signs, pedestrians, vehicles, and other important road features. Computer Vision algorithms process these images to recognize objects and understand the driving environment. The camera system helps the vehicle make accurate driving decisions in real time.

6.4 Radar Sensor

Radar sensors use radio waves to detect the speed, distance, and movement of surrounding objects. Unlike cameras, radar performs reliably in rain, fog, dust, and other challenging weather conditions. It provides additional information that improves the accuracy of object detection and supports safe driving decisions.

6.5 Sensor Fusion

Sensor Fusion is the process of combining information from cameras, LiDAR, radar, GPS, and other sensors into a single, accurate representation of the surrounding environment. This technique improves the reliability of the autonomous driving system by reducing errors and providing a complete understanding of road conditions. Sensor Fusion enables the Waymo Driver to make better decisions even when one sensor has limited visibility.

6.6 AI Computing Unit

The onboard AI computing unit acts as the brain of the autonomous vehicle. It processes the enormous amount of data collected by the sensors, runs Machine Learning and Deep Learning models, predicts the movement of surrounding objects, and determines the safest driving action. High-performance computing enables the Waymo Driver to make decisions within milliseconds.

6.7 GPS and HD Maps

The Waymo Driver uses **Global Positioning System (GPS)** together with **High-Definition (HD) Maps** for accurate localization and navigation. HD maps contain detailed information about road layouts, lane markings, traffic signs, intersections, and speed limits. By comparing live sensor data with HD maps, the vehicle can determine its exact position and follow the safest route.

6.8 Vehicle Control System

The Vehicle Control System converts AI decisions into physical vehicle actions. Based on the planned route and surrounding traffic conditions, it controls the steering wheel, brakes, and accelerator. This enables the autonomous vehicle to maintain lane position, avoid obstacles, follow traffic rules, and reach the destination safely.

Products Used in the Waymo Driver:

Product	Function
Waymo Driver	Autonomous driving system
LiDAR	Creates a 3D map of the environment
Cameras	Detect lanes, traffic signs, and objects
Radar	Measures speed and distance of nearby objects
Sensor Fusion	Combines data from all sensors
AI Computing Unit	Processes sensor data and makes decisions
GPS & HD Maps	Provides localization and navigation
Vehicle Control System	Controls steering, braking, and acceleration

WORKING MECHANISM

The 6th-Generation Waymo Driver uses Artificial Intelligence to perform autonomous driving by continuously collecting information from the surrounding environment, processing it, and making safe driving decisions in real time. The system follows a sequence of intelligent operations that enable the vehicle to travel safely without continuous human intervention.

Road Environment

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Cameras + LiDAR + Radar + GPS

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Sensor Data Collection

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Sensor Fusion

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Environment Perception

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Object Detection & Classification

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Behavior Prediction

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Path Planning

↓

Decision Making

↓

Vehicle Control

↓

Safe Autonomous Driving

Step 1: Data Collection

The autonomous vehicle continuously gathers information from cameras, LiDAR, radar, GPS, and other sensors. These sensors capture images, measure distances, detect moving objects, identify road markings, and determine the vehicle's location. The collected data provides a complete view of the surrounding environment.

Step 2: Sensor Fusion

The information collected from different sensors is combined using **Sensor Fusion**. This process merges data from cameras, LiDAR, radar, and GPS to create a single, accurate representation of the road environment. Sensor Fusion improves reliability because each sensor complements the others under different driving conditions.

Step 3: Environment Perception

Artificial Intelligence analyzes the fused sensor data to understand the surroundings. The system identifies roads, lanes, traffic lights, traffic signs, pedestrians, cyclists, vehicles, and obstacles. Computer Vision and Deep Learning models enable the vehicle to recognize and classify different objects accurately.

Step 4: Object Detection and Behavior Prediction

After identifying nearby objects, the AI predicts their future movements. For example, it estimates whether a pedestrian may cross the road, a vehicle may change lanes, or a cyclist may turn. Predicting future behavior allows the autonomous vehicle to prepare safe driving actions in advance.

Step 5: Path Planning

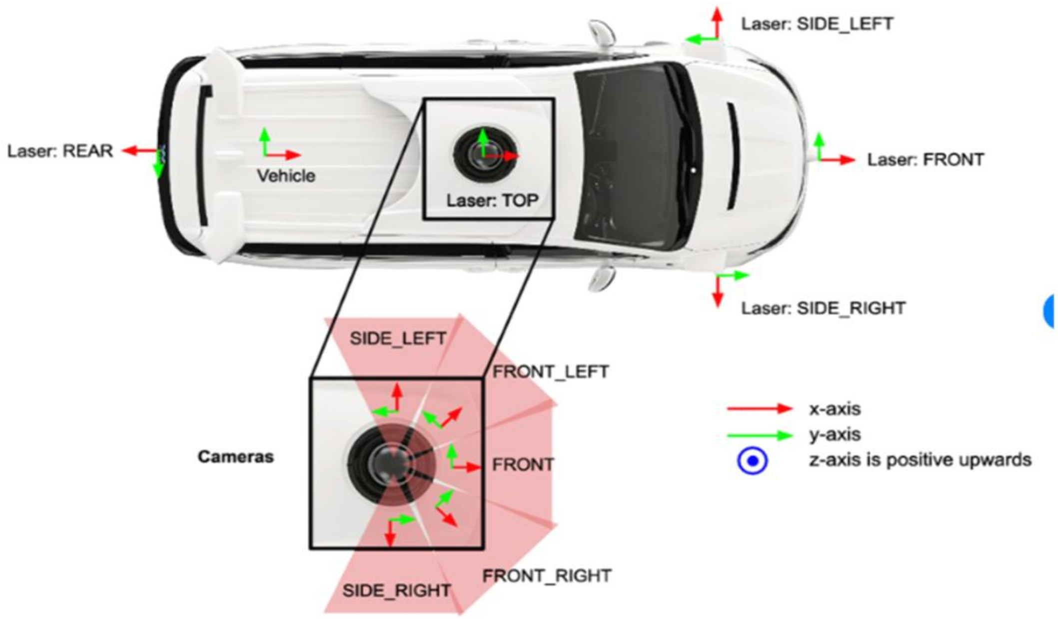
Based on the predicted movements of surrounding objects, the AI calculates the safest and most efficient driving path. It determines the best lane, speed, and direction while avoiding obstacles and following traffic rules. The planned path is continuously updated as road conditions change.

Step 6: Decision Making

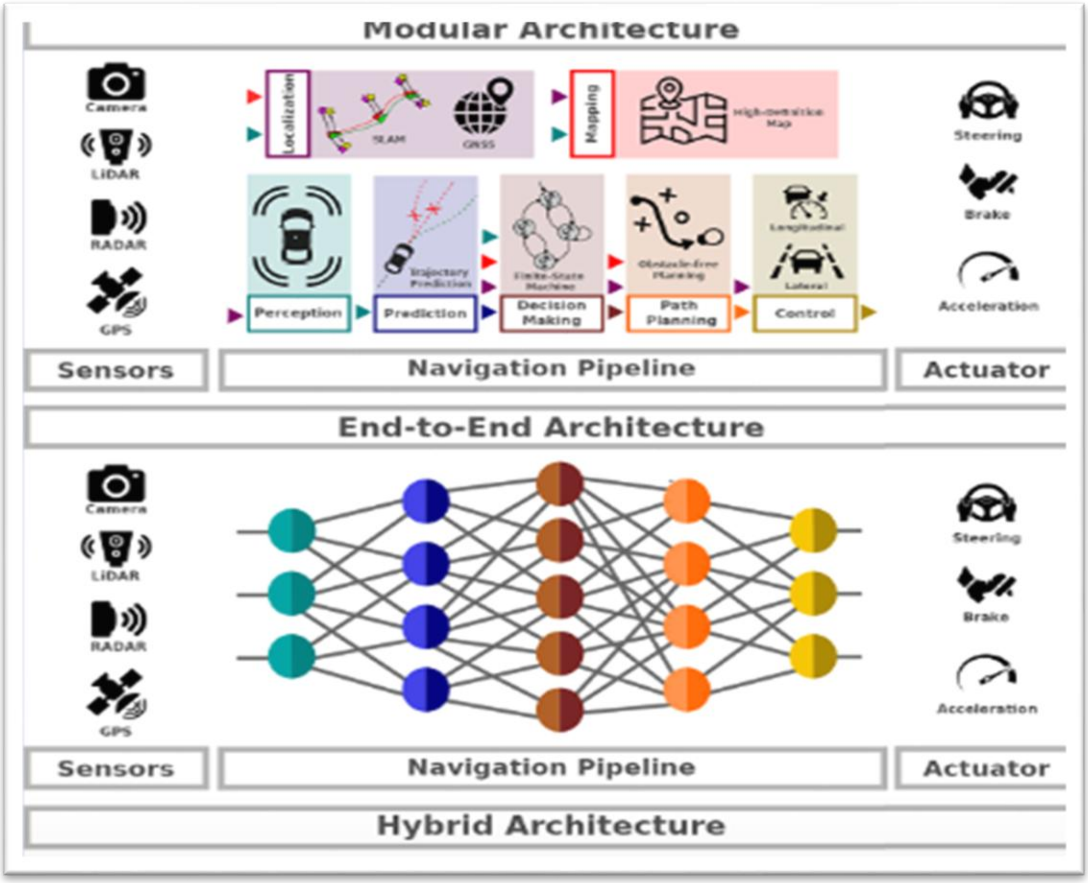
The decision-making module selects the most appropriate driving action based on the planned path. It decides whether the vehicle should accelerate, slow down, stop, change lanes, or turn. These decisions are made within milliseconds to ensure safe navigation.

Step 7: Vehicle Control

The final decisions are sent to the vehicle control system. The steering, braking, and acceleration mechanisms execute the commands generated by the AI. The vehicle continuously monitors the environment and repeats the entire process throughout the journey, enabling smooth and safe autonomous driving.



Cameras and LiDAR sensors deployed on the Waymo autonomous vehicle (Sun et al., 2020a)



DATA SOURCES AND PROCESSING

The **6th-Generation Waymo Driver** relies on multiple sources of data to understand its surroundings and make intelligent driving decisions. The autonomous vehicle continuously collects information from various sensors, including cameras, LiDAR, radar, GPS, and onboard monitoring systems. These sensors generate a large amount of real-time data, which is processed using Artificial Intelligence techniques to ensure safe and efficient driving.

The first stage of data processing is **data collection**, where cameras capture images of the road, LiDAR creates a three-dimensional map of the environment, radar measures the speed and distance of surrounding objects, and GPS provides accurate location information. Together, these sensors provide a complete understanding of the vehicle's surroundings.

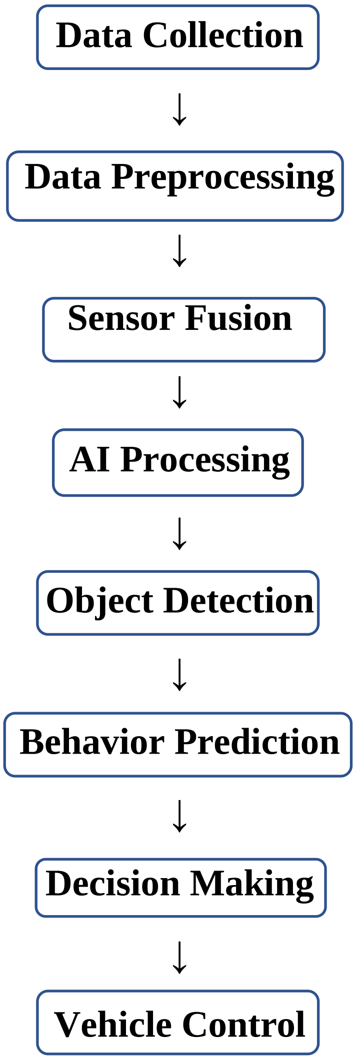
After collecting the data, the system performs **data preprocessing**. During this stage, unwanted noise and unnecessary information are removed, and the data from different sensors is synchronized. This improves data quality and prepares it for further analysis.

The processed data is then combined using **Sensor Fusion**. Sensor Fusion integrates information from cameras, LiDAR, radar, and GPS into a single, reliable representation of the environment. This enables the vehicle to accurately detect roads, traffic signals, pedestrians, cyclists, vehicles, and other obstacles even in complex driving conditions.

Data Sources Used

Data Source	Purpose
Cameras	Capture road images, traffic signs, lane markings, and pedestrians
LiDAR	Creates a 3D map of the surroundings
Radar	Detects the speed and distance of nearby objects
GPS	Determines the vehicle's location and navigation
HD Maps	Provides detailed road information for localization
Vehicle Sensors	Monitor steering, braking, speed, and vehicle status

Data Processing Pipeline



REAL-WORLD IMPLEMENTATION

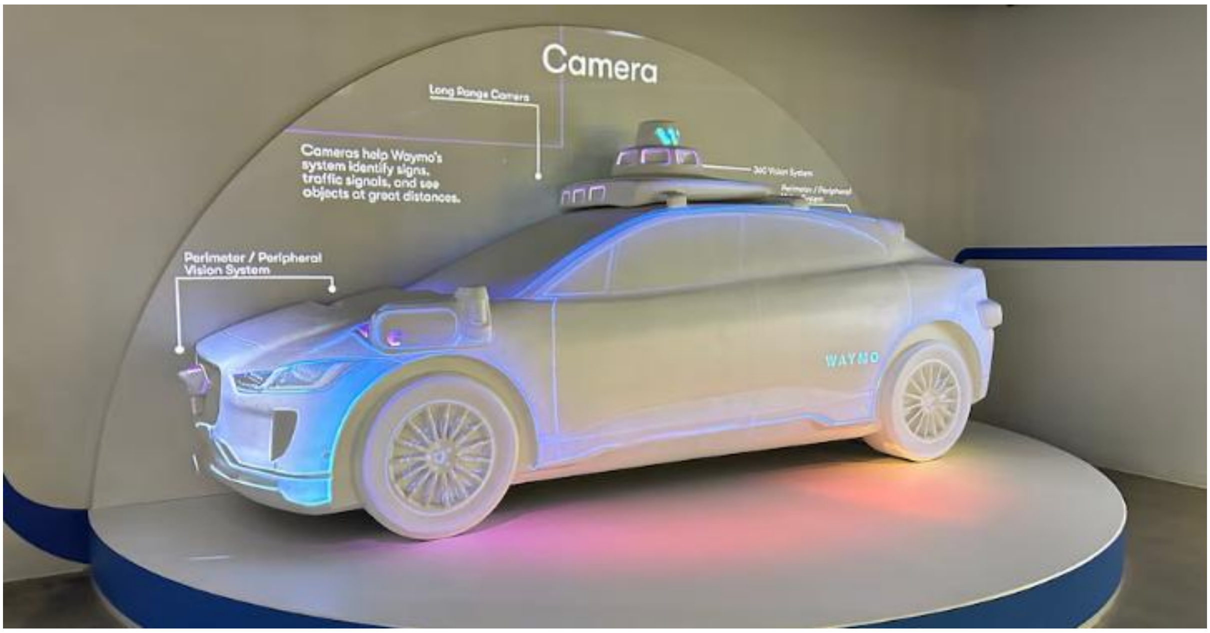
The **6th-Generation Waymo Driver** has been deployed in real-world environments to demonstrate the practical application of Artificial Intelligence in autonomous transportation. Waymo operates autonomous ride-hailing services where passengers can travel without a human driver in selected cities. The system has undergone extensive testing on public roads and has accumulated millions of kilometers of autonomous driving experience, helping improve its safety and reliability.

Waymo collaborates with automobile manufacturers to integrate the Waymo Driver into modern vehicles. The autonomous driving system has been deployed in electric vehicles equipped with advanced sensors and AI computing platforms. These vehicles can detect surrounding traffic, follow road rules, avoid obstacles, and safely transport passengers.

The real-world implementation of the Waymo Driver demonstrates how Artificial Intelligence can improve transportation efficiency, reduce human error, and support the development of smart mobility solutions. Continuous software updates and machine learning improvements allow the system to adapt to new traffic situations and enhance its overall performance over time.

Major Areas of Implementation

- Autonomous Ride-Hailing Services
- Urban Transportation
- Smart City Mobility
- Passenger Transportation
- AI Research and Autonomous Driving Development



IMPACT AND BENEFITS

The **6th-Generation Waymo Driver** has significantly advanced autonomous vehicle technology by improving road safety, transportation efficiency, and user convenience. By combining Artificial Intelligence with advanced sensing technologies, the system can make accurate driving decisions and respond quickly to changing road conditions.

One of the major benefits of the Waymo Driver is its ability to reduce human errors, such as distracted driving, fatigue, and delayed reactions, which are common causes of road accidents. The AI system continuously monitors the surroundings, detects potential hazards, and makes safe driving decisions within milliseconds.

The Waymo Driver also improves transportation efficiency by selecting optimal routes, maintaining safe distances from other vehicles, and smoothly navigating traffic. These capabilities can reduce travel time, improve traffic flow, and support more efficient urban transportation.

Another important benefit is the enhancement of passenger comfort and accessibility. Autonomous vehicles can provide transportation for elderly people, individuals with disabilities, and others who may not be able to drive independently. Continuous AI learning also allows the system to improve its performance over time through software updates and data analysis.

Benefits of the Waymo Driver

- Improves road safety by reducing human error.
- Provides real-time decision-making using AI.
- Detects pedestrians, cyclists, and obstacles accurately.
- Enhances transportation efficiency and traffic management.
- Supports autonomous ride-hailing services.
- Reduces driver fatigue during long journeys.

Comparison Table

Traditional Driving	Waymo Driver
Human-controlled driving	AI-controlled autonomous driving
Human reaction time	Millisecond AI response
Driver fatigue	No fatigue during operation
Limited observation	360° sensor monitoring
Manual navigation	Intelligent route planning
Higher possibility of human error	AI-assisted decision-making

CHALLENGES AND LIMITATIONS

Although the **6th-Generation Waymo Driver** represents a major advancement in autonomous vehicle technology, several challenges remain before widespread adoption can be achieved.

One of the biggest challenges is operating safely in highly complex and unpredictable traffic situations. Construction zones, emergency vehicles, unusual road layouts, and unexpected pedestrian behavior require advanced AI reasoning and careful system validation.

Weather conditions such as heavy rain, fog, snow, and dust can affect the performance of cameras and LiDAR sensors. Although multiple sensors improve reliability, extreme environmental conditions can still reduce sensing accuracy.

Cybersecurity is another important concern. Since autonomous vehicles rely heavily on software, communication systems, and onboard computers, they must be protected against unauthorized access and cyberattacks.

Legal and ethical issues also present challenges. Governments must develop regulations for autonomous driving, define liability in case of accidents, and establish safety standards. Public trust is equally important, as many people are still hesitant to rely completely on self-driving vehicles.

In addition, the development and deployment of autonomous vehicle technology require significant investment in AI research, sensor hardware, computing systems, and testing infrastructure.

Major Challenges

- Complex traffic environments
- Adverse weather conditions
- Sensor limitations
- Cybersecurity risks
- Legal and regulatory challenges
- Ethical decision-making
- Public acceptance
- High development and maintenance costs

FUTURE SCOPE AND EMERGING TRENDS

Artificial Intelligence is expected to continue transforming autonomous transportation in the coming years. Future versions of the Waymo Driver are likely to become more intelligent, efficient, and capable of handling increasingly complex driving environments.

Advancements in Deep Learning, Computer Vision, and Machine Learning will improve object detection, behavior prediction, and decision-making accuracy. Future autonomous vehicles may operate safely under a wider range of weather and road conditions while reducing computational delays.

The integration of autonomous vehicles with **Smart Cities**, **5G communication**, **Vehicle-to-Vehicle (V2V)** communication, and **Vehicle-to-Infrastructure (V2I)** communication is expected to improve traffic management and road safety. Future systems may also support better coordination between vehicles and traffic signals, reducing congestion and improving travel efficiency.

Researchers are also working on improving AI explainability, cybersecurity, energy efficiency, and ethical decision-making. Continuous software updates and large-scale real-world data collection will further enhance the safety and reliability of autonomous driving systems.

Future Developments

- Smarter AI algorithms
- Better Computer Vision models
- Improved sensor technologies
- Integration with Smart Cities
- Vehicle-to-Vehicle (V2V) communication
- Vehicle-to-Infrastructure (V2I) communication
- Enhanced cybersecurity
- Wider commercial deployment

CONCLUSION

The **6th-Generation Waymo Driver** represents one of the most advanced applications of Artificial Intelligence in autonomous vehicles. By combining Machine Learning, Deep Learning, Computer Vision, Sensor Fusion, and intelligent decision-making, the system enables vehicles to perceive their surroundings, analyze traffic conditions, and drive safely with minimal human intervention.

The study demonstrates how AI can improve road safety, transportation efficiency, and passenger convenience while reducing the impact of human error. Real-world deployment of autonomous vehicles shows the growing potential of AI in creating intelligent transportation systems.

Although challenges such as adverse weather conditions, cybersecurity, regulations, and public acceptance remain, continuous research and technological advancements are expected to address these issues. As AI technology continues to evolve, autonomous vehicles will play an increasingly important role in building safer, smarter, and more sustainable transportation systems.

In conclusion, the **6th-Generation Waymo Driver** highlights the significant contribution of Artificial Intelligence to the future of mobility. Its ability to perceive, predict, plan, and make intelligent driving decisions demonstrates how AI is transforming modern transportation and paving the way for fully autonomous mobility solutions.

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